CS257 Linear and Convex Optimization

Homework 9

Due: November 16, 2020

November 23, 2020

For this assignment, you should submit a report in pdf format as well as your source code (.py files). The report should include all necessary figures, the outputs of your Python code, and your answers to the questions. Do NOT write your answers in any of the .py files.

1. Consider the optimization problem $\min_x f(x)$, where $f : \mathbb{R} \to \mathbb{R}$ is given by $f(x) = (x-a)^4$, and $a \in \mathbb{R}$ is a constant.

(a). Find an explicit expression for the Newton step.

(b). Let x_k be the sequence of iterates generated by Newton's method. Let $y_k = |x_k - a|$ be the error between the k-th iterate and the optimal solution. Show that $y_{k+1} = \frac{2}{3} y_k$

(c). Conclude $|x_k - a|$ decays to zero exponentially, i.e. x_k converges exponentially to a.

Proof. (a). $f'(x) = 4(x-a)^3$, $f''(x) = 12(x-a)^2$.

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)} = x_k - \frac{4(x_k-a)^3}{12(x_k-a)^2} = x_k - \frac{1}{3}(x_k - a) = \frac{2}{3}x_k + \frac{1}{3}a$$

(b). Since

$$x_{k+1} = x_k - \frac{1}{3}(x_k - a)$$

we obtain

$$x_{k+1} - a = (x_k - a) - \frac{1}{3}(x_k - a) = \frac{2}{3}(x_k - a)$$

Taking the absolute values,

$$y_{k+1} = \frac{2}{3} y_k$$

(c).

$$|x_k - a| = y_k = \left(\frac{2}{3}\right)^k y_0$$

converges exponentially to 0.

□

Remark. Note that the convergence rate here is only exponential no matter how close x_0 is to $x^* = a$, while the rate given by the theorem in Lecture 10 is doubly exponential, which is much faster, at least when x_0 is close enough to x^*. This is because $(x-a)^4$ is not strongly convex, which does not satisfy the assumptions of the theorem.

2. Consider $\min_x f(x)$ for $f(x) = x - \log x$, where log denotes the natural logarithm. This is an unconstrained problem, but the domain of f is dom $f = (0, +\infty)$.

First implement the function `newton` and `damped_newton` in `newton.py`, then complete the code in `hw9.py`.

(a). Find the exact optimal solution and optimal value analytically.

(b). Find an explicit expression for the Newton step.

(c). Implement and run the **Newton's** method in for 5 iterations, using the initial points $x_0 = 0.5$, 1.5 and 2.5. Show the values of x_k for each iteration. What do you observe?

(d). Let $y_k = x^* - x_k$, where x^* is the optimal solution you find in (a). Show $y_{k+1} = y_k^2$ and hence $y_k = y_0^{2^k}$. Conclude that $x_k \to x^*$ if and only if $|x_0 - x^*| < 1$. Is this consistent with what you see in (c)?

(e). Implement and run the **damped Newton's** method using the same initial points $x_0 = 0.5$, 1.5 and 2.5, with $\alpha = 0.1$ and $\beta = 0.7$. Note that you should make sure the point is in the domain before using armijo condition (Hint: you can add the domain check to the WHILE condition for the inner loop in `newton.py` or let function f(x) in `hw9.py` return positive infinity if the point is not in the domain). Show the values of x_k, plot the error $f(x_k) - f(x^*)$ and the step sizes for each iteration. Compare Newton's method with damped Newton's method basing on the results you obtain in (c) and (e).

Proof. (a). f is convex. Setting $f'(x) = 1 - \frac{1}{x} = 0$, we find $x^* = 1$ and $f(\boldsymbol{x}^*) = 1$.

(b). $f''(x) = \frac{1}{x^2}$. The Newton step is

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)} = x_k - \frac{1 - 1/x_k}{1/x_k^2} = 2x_k - x_k^2$$

(c).
```
x0 = 0.5
x1 = 0.75
x2 = 0.9375
x3 = 0.99609375
x4 = 0.9999847412109375
x5 = 0.9999999997671694

x0 = 1.5
x1 = 0.75
x2 = 0.9375
x3 = 0.99609375
x4 = 0.9999847412109375
x5 = 0.9999999997671694

x0 = 2.5
x1 = -1.25
```

```
x2 = -4.0625
x3 = -24.62890625
x4 = -655.8408355712891
x5 = -431438.8832739892
```

For $x_0 = 0.5, 1.5$, $x_k \to x^*$. For $x_0 = 2.5$, x_k goes outside the domain of f and diverges.

(d). By (b),

$$1 - x_{k+1} = 1 - 2x_k + x_k^2 = (1 - x_k)^2$$

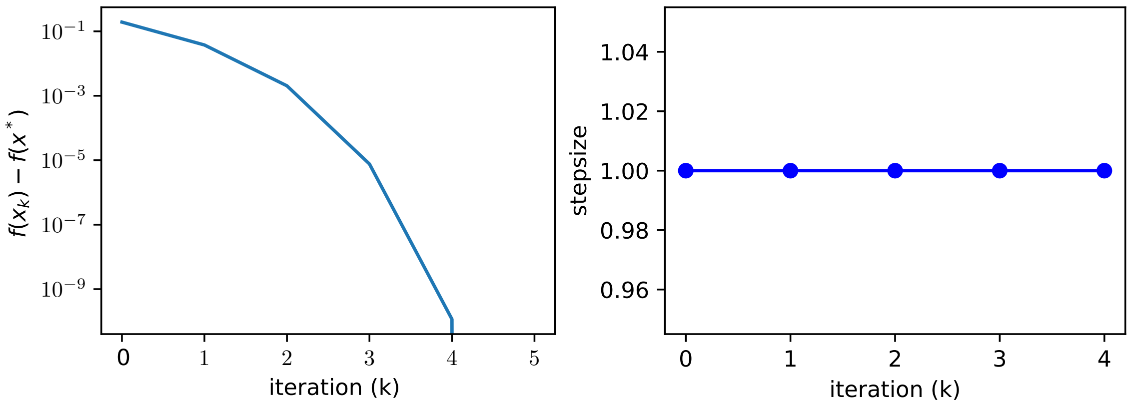
i.e. $y_{k+1} = y_k^2$. By induction, $y_k = y_0^{2^k}$, which goes to 0 iff $|y_0| < 1$. Thus $x_k \to x^* = 1$ iff $|x_0 - 1| < 1$ or $x_0 \in (0, 2)$. This is consistent with the observations in (c).

(e).
```
x0 = 0.5
x1 = 0.75
x2 = 0.9375
x3 = 0.99609375
x4 = 0.9999847412109375
x5 = 0.9999999997671694

x0 = 1.5
x1 = 0.7499999999999999
x2 = 0.9375
x3 = 0.99609375
x4 = 0.9999847412109375
x5 = 0.9999999997671694

x0 = 2.5
x1 = 0.6625000000000003
x2 = 0.8860937500000001
x3 = 0.9870253662109376
x4 = 0.9998316588780397
x5 = 0.9999999716612668
```

From the results we can see that damped Newton's method converges even for $x_0 = 2.5$. The step sizes used by damped Newton's method show that in pure Newton phase backtracking always selects step size 1, and damped Newton's method is the same as Newton's method in this phase.

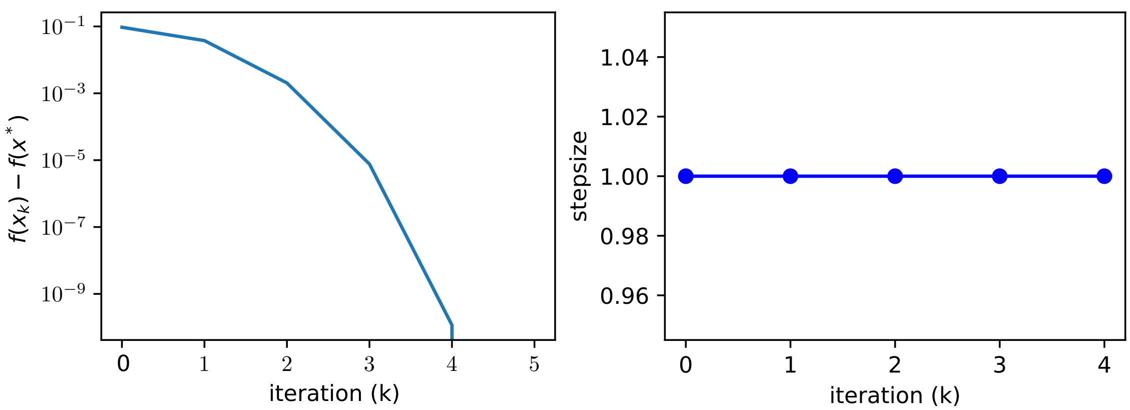
□

Remark. As we mentioned in the lecture, Newton's method may diverge when $\boldsymbol{x}_0$ is far away from $\boldsymbol{x}^*$. A remedy is provided by damped Newton's method, which guarantees global convergence, but there is no guarantee that the convergence rate is doubly exponential globally. By contrast, damped Newton's method can guarantee global convergence.



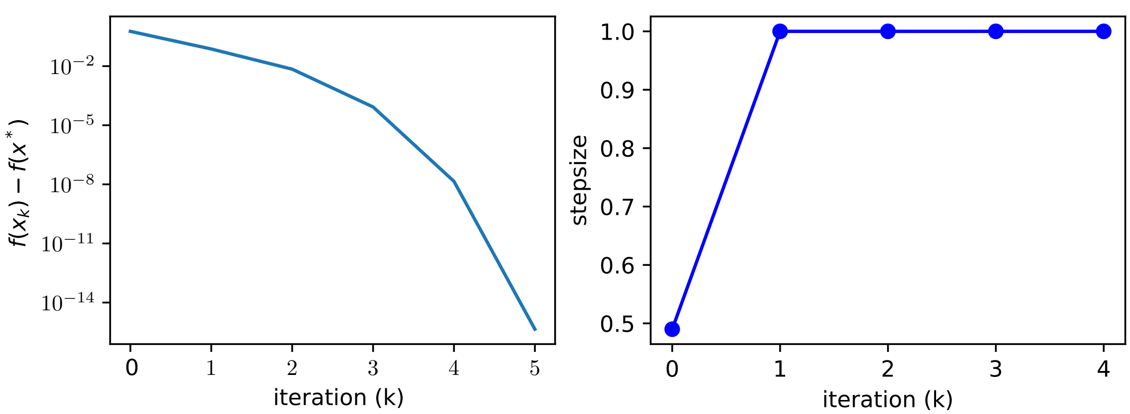
(a) $f(x_k) - f(x^*)$ **(b)** *Stepsizes*

Figure 1: $x_0 = 0.5$



(a) $f(x_k) - f(x^*)$ **(b)** *Stepsizes*

Figure 2: $x_0 = 1.5$



(a) $f(x_k) - f(x^*)$ **(b)** *Stepsizes*

Figure 3: $x_0 = 2.5$